A Full Stack Development Project Report

ON

FOODBRIDGE: A Smart Food Donation and Distribution Management System

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1. Title of the Project

FoodBridge: An Intelligent Food Donation & Distribution Management System

2. Reason for Choosing the Topic

The majority of us must have come across a wedding ceremony or a mess hall where a pile of perfectly edible food gets thrown away. Meanwhile, there are people living on the other side of the street who go hungry every day. While investigating the reason behind the occurrence of such an issue, we realized that the main cause is communication. A restaurant would be happy to donate the food, but it finds itself in no position to get in touch with an NGO before it gets spoiled and, on the other hand, an NGO knows nothing of that food.

We decided to choose this topic because we wanted our DBMS project to be relevant rather than some another hospital or library management system. Besides, the topic was quite suitable for the subject, as all that it requires is to keep the proper record of donations and their distribution.

3. Objectives of the project

- Creation of an integrated application where the donor will be able to post, validate, request, allocate and distribute the surplus food.
- Allowing role based access, i.e., donor, NGO/volunteer and administrator have access to their own views separately.
- Prevention of errors such as over-allocation of the food through constraints, triggers, and validations.
- Save all the donations in different statuses (pending, approved, distributed).
- Integration of both web programming and database technologies into one system.

4. Customers/Target audience

- The Donor - Restaurant owner, grocery stores, hostels, hotels, event planners and individual homes having surplus food.
- The NGO/volunteer - NGOs authorized to pick up the donations and distribute them among needy ones.
- The Administrator – Coordinator who is responsible for validation, approval and distribution of food donations.
- The Beneficiary (Indirectly) – Shelters, community centers and individual people who receive the food donation.

5. Project Summary

This project is a web application created in PHP and MySQL with HTML, CSS, Bootstrap and JavaScript used in the front-end. Also, there is a desktop Java Swing application with JDBC connection that provides access to the same data for staff members. The authentication uses email and password. The role of the session defines what pages a certain user can use (Admin, Donor or NGO_Volunteer).

Key Features. The donors add food items that include the category of food, quantity, preparation and expiration date and the address where the items can be picked up. Administrators check those food items and approve them. NGOs find the approved food items and make a food request that includes the quantity of food, the number of beneficiaries and the address for delivery. Administrators make allocation of stock among the food requests. The delivery team tracks the time of dispatch and delivery of the food item. The dashboard gives the statistical information about the total number of donations, pending requests, approved food items, completed deliveries and the total amount of food.

Database schema. There is a MySQL database FoodDonationDistributionDB with nine tables: USER, ADMIN, DONOR, NGO_VOLUNTEER, FOOD_CATEGORY, DONATION, FOOD_REQUEST, ALLOCATION and DISTRIBUTION.

6. Git Repository Link

https://github.com/karanpalbtech2025-kkp/Food-Donation.git